

NGUYEN HUU HOAI

AI Engineer

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PROFESSIONAL SUMMARY

AI Engineer graduate with hands-on experience in NLP, Computer Vision, and LLM integration. Built production-ready AI pipelines at FPT Software using Azure OpenAI, and developed end-to-end CV parsing systems combining YOLO layout detection with LLM-based extraction. Strong communicator with experience presenting research and collaborating in cross-functional teams. Eager to contribute to international projects and grow in a professional AI engineering environment.

TECHNICAL STACKS

- **Languages:** Python, SQL, C#
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO
- **Version Control:** Git, Gitlab
- **Backend & APIs:** RESTful APIs, .NET Core
- **Database:** CosmosDB, PostgreSQL
- **Cloud & AI Platforms:** Azure OpenAI, Azure AI Services

EDUCATION BACKGROUND

- Bachelor of Engineering in Artificial Intelligence – FPT University, Quy Nhon AI Campus
GPA: 3.3/4.0

LANGUAGE SKILLS

- English: Intermediate (B2) – written and verbal communication for technical contexts
- Japanese: Basic

PROFESSIONAL EXPERIENCE HIGHLIGHTS (FPT SOFTWARE)

FPT Software

Project: Estimation Tool (05/2025 - 02/2026)

An AI-powered tool that assists Project Managers in estimating task effort points by analyzing sprint inputs through an LLM pipeline via Azure OpenAI.

Role: AI & Backend Developer (Intern)

Responsibilities:

- Engineered an LLM pipeline via Azure OpenAI (GPT) to automate sprint effort estimation for Project Managers, reducing manual estimation effort in the delivery workflow.
- Designed and iterated prompt engineering strategies (system prompts, context formatting, output schema enforcement) through 25 evaluation cycles to improve output consistency and PM acceptance rate.
- Processed and structured sprint input data into a clean context pipeline for the LLM, with output parsing and quality validation logic built in Python.

Technologies Used:

- Python, C#
- Sklearn, Pandas, .Net Core, CosmosDB
- Azure AI Foundry, Azure OpenAI
- Gitlab, HAPI

OTHER EXPERIENCE

Project: Fipilot – AI-Powered Resume Analysis Pipeline (05/2026 – Present)

An end-to-end document AI system that parses Vietnamese CVs using YOLO-based layout detection, PyMuPDF text extraction, and LLM-based structured data extraction via Qwen/Llama.

Role: AI Engineer**Responsibilities:**

- Designed and trained YOLOv12s/YOLO26s models for CV layout detection, achieving best-in-class performance across 5 model iterations.
- Built a fine-tuned SLM extraction pipeline (Qwen3-0.6B via LoRA/SFT) using Jinja2 prompt templates and Pydantic/BAML schema constraints to parse structured JSON from Vietnamese resume text.
- Developed a hybrid interview engine combining template-based questioning with LLM-driven adaptive paraphrasing – bot reads from a curated Q&A template (~60%) while contextualizing questions to the candidate's CV projects (~40%).

Technologies Used:

- Python, PyTorch, YOLO (v12s, 26s), PyMuPDF
- Qwen(SLM), Jinja2, Pydantic
- Git, uv, NumPy, OpenCV

Project: Invoice OCR System (01/2025 – 05/2025)

Research project tackling Vietnamese invoice OCR – a domain with high complexity due to mixed fonts, layouts and handwriting – benchmarked against MC-OCR competition solutions.

Role: AI Engineer / Research Developer**Responsibilities:**

- Reduced invoice OCR Character Error Rate by 22%, benchmarked against competition-tier baselines.

- Engineered multi-model pipeline (YOLOv11 → VietOCR → PhoBERT) with controlled fine-tuning on Vietnamese invoice datasets.
- Built systematic evaluation framework with bounding-box visualization for iterative model comparison.

Technologies Used:

- Python, PyTorch, TensorFlow
- YOLOv11, PhoBERT, VietOCR
- NumPy, OpenCV

Project: Scene Text Recognition (05/2024 – 11/2024)

A computer vision system that automatically detects and recognizes text in natural scenes – such as street signs and advertisements – using deep learning.

Role: AI Developer

Responsibilities:

- Developed a scene text recognition system for natural environments (street signs, advertisements) using deep learning, delivering accurate text extraction via a web-based interface.
- Engineered a full-stack pipeline integrating YOLOv11 detection and ResNet + Bi-LSTM + CTC architecture, with OpenCV preprocessing to handle real-world noise (perspective distortion, lighting variance).

Technologies Used:

- Python, TensorFlow, YOLOv11
- OpenCV, NumPy, Scikit-learn